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Large Eddy Simulation with the Immersed Boundary Method: Implementation of an adaptive wall-modeling approach for complex turbulent flows

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Abstract

High-Reynolds number turbulent flows dominate critical engineering applications, ranging from aircraft and other vehicles to wind turbines and turbomachinery. However, their simulation remains an open challenge: such flows are characterized by a multi-scale nature, resulting in an enormous number of degrees of freedom; consequently, the computational cost of resolving all scales via Direct Numerical Simulation (DNS) becomes prohibitive. Large Eddy Simulation (LES) addresses this problem by resolving the largest scales of the motion while modeling the smaller ones. Furthermore, when dealing with complex geometries, the Immersed Boundary Method (IBM) provides a flexible framework that avoids the complexity of body-fitted meshes. However, the need to resolve the boundary layer still imposes strict grid requirements, driving the adoption of Wall-Modeled LES (WM-LES).

Standard wall models typically rely on the equilibrium boundary layer assumption, which neglects the pressure gradient and convective terms and employs an equilibrium formulation for the turbulent viscosity. While computationally efficient, these models often fail to predict complex phenomena such as flow separation and reattachment. Conversely, with a higher computational effort, non-equilibrium models based on Ordinary or Partial Differential Equations (ODEs/PDEs) offer greater accuracy.

In the current work, the implementation of an adaptive wall modeling approach is proposed. The aim is to optimize the trade-off between cost and accuracy by dynamically selecting the governing equations for the wall stress, enabling a more accurate ODE model only in regions where the flow complexity requires it, while maintaining the efficiency of standard approaches in the regions where the equilibrium boundary layer conditions can be considered valid.

Nomenclature

Symbol	Description
x_i, x	Spatial component
t	Time
p	Pressure
$\boldsymbol{e}_{ij}$	Strain rate tensor
$\boldsymbol{\omega}_{ij}$	Rotation rate tensor
μ	Dynamic viscosity
ν	Kinematic viscosity
ρ	Fluid density
Re	Reynolds number
$\boldsymbol{f}$	Body forces
u_i, u	Instantaneous velocity component
U, $\langle u \rangle$	Mean flow velocity, Reynolds Averaged
u'	Velocity fluctuation, Reynolds decomposition
$\widetilde{u}_i$, $\widetilde{u}$	Filtered velocity, LES Resolved field
u''	Residual component, LES Sub-Grid Scale
$\boldsymbol{F}_{\mathcal{K}}$, $\boldsymbol{F}_k$	Flux of kinetic energy for mean and turbulent flow
$\mathcal{K}$, k	Kinetic energy of the mean and turbulent flow
$\mathcal{P}$	Turbulent energy production
$\mathcal{D}$	Dissipation for the mean flow
$\mathcal{E}$	Dissipation for the turbulent flow
Ω	Generic large integration control volume
$\mathbb{C}_{\mathcal{K}}$, $\mathbb{C}_k$	Convection terms for mean and turbulent flow, integrated over Ω
$\mathbb{F}_{\mathcal{K}}$, $\mathbb{F}_k$	Flux of kinetic energy for mean and turbulent flow, integrated over Ω
$\mathbb{D}_{\mathcal{K}}$	Dissipation for the mean flow, integrated over Ω
$\mathbb{P}$	Turbulent energy production, integrated over Ω
$\mathcal{E}$	Dissipation for the turbulent flow, integrated over Ω

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Table 1: Nomenclature (continued)

Symbol	**Description**
L_0, U_0	Characteristic length and velocity scales of the domain
ℓ_0, u_0, τ_0	Characteristic length, velocity and time of the large scales
η, u_η, τ_η	Characteristic length, velocity and time of the Kolmogorov scales
τ	Total shear stress
τ_W	Wall shear stress
P_E	Mean free-stream pressure
P_W	Mean wall pressure
h	Channel half-height
U_C	Centerline mean velocity
u_τ	Friction velocity
δ_ν	Viscous length scale
U^+	Non-dimensional mean velocity (inner units)
y^+	Non-dimensional wall distance (inner units)
Re_τ	Friction Reynolds number
κ	Von Kármán constant
B	Log-law integration constant
G	Spatial filter kernel function
Δ	Grid spatial resolution
Δ_F	Filter width
τ_{ij}^R	Residual (Sub-Grid Scale) stress tensor
k_R	Residual (Sub-Grid Scale) flow kinetic energy
ν_R	Residual (Sub-Grid Scale) eddy viscosity
C_S	Tuning constant for Smagorinsky model
$\mathcal{S}_{ij}$	Traceless symmetric part of the square of the velocity gradient tensor
C_W	Tuning constant for WALE model
$\mathcal{AD}$, $\mathcal{AD}_k$	Advective/diffusive term (at sub-step k)
$\boldsymbol{u}_n$, $\boldsymbol{u}_k$	Velocity field at step n, at sub-step k
α_k, β_k, γ_k	Coefficients for the low storage Runge-Kutta scheme
$\boldsymbol{u}^*$	Prediction velocity field
p^*	Correction pressure field
CFD	Computational Fluid Dynamics
DNS	Direct Numerical Simulation
IBM	Immersed Boundary Method

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Table 1: Nomenclature (continued)

Symbol	**Description**
LES	Large Eddy Simulation
RANS	Reynolds-Averaged Navier-Stokes
SGS	Sub-Grid Scale
SST	Shear Stress Transport
URANS	Unsteady RANS
WM-LES	Wall-Modeled LES
WR-LES	Wall-Resolved LES

Chapter 1

Introduction

1.1 Computational Fluid Dynamics

Computational Fluid Dynamics (CFD) is the study of fluids through a numerical computer-based approach, using algorithms to find solutions to the equations that describe the fluid behavior. The information produced have mainly two purposes: in the field of research, it allows a better understanding of the physical mechanisms governing the fluid motion, and the consequent modelling and control of such mechanisms. In the engineering analyses, fluid simulation is used as a prediction tool, especially during the product design phase, to assess flow characteristics and physical parameters, such as lift and drag forces generated, propagation of acoustic waves or convective heat transfer.

As all mathematical models, the definition of the level of approximation have a primary importance also for fluid simulation. This choice concern various aspects of the description of the phenomena. Firstly the scale of reality considered: the reality can be described at an atomic, particle or micro- and macroscopic level. The last one is that used by classical mechanics, in particular continuum mechanics, that is the case considered in CFD.

The second aspect to be defined in terms of level of approximation is the accuracy of dynamic description. It consist on the choice of which forces should be considered exerted on the system, based on their relevance. The most general case is represented by the Navier-Stokes equations complemented by constitutive laws (empirical laws characterizing the fluid and state laws). This case can be simplified, obtaining for example the Euler equations, for the description of inviscid flows.

Another aspect regards the scale separation problem, in particular the spatial and temporal resolution of choice. To obtain a description of maximum quality, all the space-time scales of the solution must be taken into account while solving the unsteady Navier-Stokes

equations. As will be discussed later, it turns out that such a solution criterion is extremely restrictive for multi-scale phenomena, like in the case of turbulent flows.

Chapter 2

Turbulent flows

In most of the engineering applications, flows are prevalently turbulent. Generally this is the case of fluid processed by pumps, compressors or flows around cars, trains, airplanes and many more situations in the everyday surrounding.

Such flows are characterized by some specific properties. A key feature is the multi-scale nature of the vortical structures, whose dimensions span 6 to 8 orders of magnitude. Despite the deterministic nature (they can be described accurately by the Navier-Stokes equations), the strong dependence and sensitivity to initial conditions make their behavior chaotic. Furthermore, turbulent flows are intrinsically three-dimensional and unsteady, they show an high diffusivity (mixing of momentum, energy and species) and also an increased wall friction when compared to laminar flows.

The Navier-Stokes equations express the principles of mass and momentum conservation, governing the evolution of the flow field. In the current work, the fluid is assumed to be incompressible and Newtonian, with constant viscosity. Under these hypotheses, the governing equations can be formulated in their *convective form* as:

$$\begin{cases} \nabla \cdot \boldsymbol{u} = 0 \\ \dfrac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} = -\dfrac{\nabla p}{\rho} + \nabla \cdot (2\nu \boldsymbol{e}) + \boldsymbol{f} \end{cases} \tag{2.1}$$

or recasting the system in its *conservative form*:

$$\begin{cases} \nabla \cdot \boldsymbol{u} = 0 \\ \dfrac{\partial \boldsymbol{u}}{\partial t} + \nabla \cdot (\boldsymbol{u} \otimes \boldsymbol{u}) = -\dfrac{\nabla p}{\rho} + \nabla \cdot (2\nu \boldsymbol{e}) + \boldsymbol{f} \end{cases} \tag{2.2}$$

where $\boldsymbol{f}$ represents the body forces.

The flow regime is described by the Reynolds number, a non-dimensional value representing the ratio between inertial and viscous forces, and defined as:

$$Re = \frac{\rho\, U_0\, L_0}{\mu} = \frac{U_0\, L_0}{\nu} \tag{2.3}$$

where U_0 and L_0 are the characteristic velocity and length scale of the flow. It is observed that, for higher values of Re, the magnitude of velocity gradients is progressively more intense, and the scale of smaller vortices decreases.

Considering the chaotic nature of the turbulence, a deterministic and instantaneous description of the phenomena is not always the most efficient option. Consequently, the analysis is typically focused on a statistical approach, considering averaged fields. It can be observed how those averaged fields resemble qualitatively the laminar behavior, and their dependence on the Reynolds number diminishes significantly.

2.1 Reynolds-Averaged Navier-Stokes equations

To analytically formalize the idea of considering the average fields and the turbulent one separately, the Reynolds averaging operator and the Reynolds decomposition are here introduced. The former consists of an ensemble average over a number of realizations. In more practical terms this implies repeating the same experiment N times, under identical conditions, and averaging the flow variables at the exact same location and time instant across all the independent measures. Considering to apply the Reynolds operator for the velocity component u:

$$\langle u(\boldsymbol{x}, t)\rangle = U(\boldsymbol{x}, t) = \frac{1}{N} \sum_{k=1}^{N \gg 1} u_{(k)}(\boldsymbol{x}, t) \tag{2.4}$$

Instead for what concerns the Reynolds decomposition, it can be defined as:

$$u(\boldsymbol{x}, t) = U(\boldsymbol{x}, t) + u'(\boldsymbol{x}, t) \tag{2.5}$$

where u' is the fluctuation component and U is the averaged quantity.

By substituting in the Navier-Stokes equations (2.2), neglecting body forces for simplicity, the Reynolds decomposition and applying the Reynolds operator the Reynolds-Averaged Navier-Stokes (RANS) equations are obtained:

$$\begin{cases} \boldsymbol{\nabla} \cdot \boldsymbol{U} = 0 \\ \dfrac{D\boldsymbol{U}}{Dt} = -\dfrac{\boldsymbol{\nabla} P}{\rho} + \boldsymbol{\nabla} \cdot (2\nu \boldsymbol{E}) - \boldsymbol{\nabla} \cdot \langle \boldsymbol{u}'\boldsymbol{u}' \rangle \end{cases} \tag{2.6}$$

where the term $\langle \boldsymbol{u}'\boldsymbol{u}' \rangle$ is the Reynolds stress tensor, and its modelling represents the closure problem of the RANS equations.

2.2 Energy cascade

The RANS equations introduced above, allow demonstrating some fundamental mechanisms governing the energy dynamics within turbulent flows. Omitting for brevity the detailed algebraic steps, the mean and turbulent flow kinetic energies can be defined respectively as:

$$\mathcal{K} = \frac{1}{2}\boldsymbol{U}\cdot\boldsymbol{U} = \frac{1}{2}U_i\,U_i \qquad k = \frac{1}{2}\langle \boldsymbol{u}'\cdot\boldsymbol{u}' \rangle = \frac{1}{2}\langle u_i'\,u_i' \rangle \tag{2.7}$$

By manipulating the momentum equation, it is possible to derive a formulation for both the time derivative of the mean kinetic energy and turbulent kinetic energy terms:

$$\frac{D\mathcal{K}}{Dt} = \underbrace{\frac{\partial}{\partial x_j}\left[-\frac{P}{\rho}U_j + 2\nu\,E_{ij}\,U_i - \langle u_i'u_j'\rangle\,U_i\right]}_{\boldsymbol{\nabla}\cdot\boldsymbol{F}_{\mathcal{K}}} \underbrace{-2\nu\,E_{ij}E_{ij}}_{-\mathcal{D}} + \underbrace{\left(\langle u_i'u_j'\rangle\,\frac{\partial U_i}{\partial x_j}\right)}_{-\mathcal{P}} \tag{2.8}$$

$$\frac{Dk}{Dt} = \underbrace{\frac{\partial}{\partial x_j}\left[-\frac{\langle p'u_j'\rangle}{\rho} + 2\nu\,\langle e_{ij}'u_i'\rangle - \frac{\langle u_i'u_i'u_j'\rangle}{2}\right]}_{\boldsymbol{\nabla}\cdot\boldsymbol{F}_k} \underbrace{-2\nu\,\langle e_{ij}'e_{ij}'\rangle}_{-\mathcal{E}} + \underbrace{\left(-\langle u_i'u_j'\rangle\,\frac{\partial U_i}{\partial x_j}\right)}_{\mathcal{P}} \tag{2.9}$$

where $\boldsymbol{F}_{\mathcal{K}}$ and $\boldsymbol{F}_k$ are the fluxes of kinetic energy (spatial transport terms), $\mathcal{D}$ and $\mathcal{E}$ are the dissipation terms respectively for the mean and turbulent flows and $\mathcal{P}$ is the turbulent energy production. In a more compact form, the two equation can be written as:

$$\frac{D\mathcal{K}}{Dt} = \boldsymbol{\nabla}\cdot\boldsymbol{F}_{\mathcal{K}} - \mathcal{D} - \mathcal{P} \qquad \frac{Dk}{Dt} = \boldsymbol{\nabla}\cdot\boldsymbol{F}_k - \mathcal{E} + \mathcal{P} \tag{2.10}$$

Following the approach described by Pope [1], the equations can be integrated over a sufficiently large control volume Ω, surrounding a free shear flow such as a turbulent jet ($Re \gg 1$), in stationary conditions. This perspective, and what result from the integration, allows to identify which component are the dominant in the energy dynamics.

$$\int_\Omega \frac{\partial \mathcal{K}}{\partial t}\,d\Omega = \mathbb{C}_{\mathcal{K}} + \mathbb{F}_{\mathcal{K}} - \mathbb{D}_{\mathcal{K}} - \mathbb{P} \implies \mathbb{C}_{\mathcal{K}} \approx \mathbb{P} \tag{2.11}$$

$$\int_\Omega \frac{\partial k}{\partial t}\,d\Omega = \mathbb{C}_k + \mathbb{F}_k - \mathcal{E} + \mathbb{P} \implies \mathbb{P} \approx \mathcal{E} \tag{2.12}$$

The analysis of the dominant terms in the energy balance equations reveals how the input of energy, associated with convection ($\mathbb{C}_{\mathcal{K}}$) and its viscous dissipation ($\mathcal{E}$) occur at different

scales. The apparent decoupling between energy input and output is better described by Richardson's theory.

2.3 Richardson's theory

The first concept in Richardson's view of the energy cascade is that the turbulence can be considered to be composed of eddies of different sizes [1].

Considering the eddies with the largest sizes, their characteristic dimensions can be evaluated as: $\ell_0 \approx L_0$ and $u_0 \approx u'_{rms} = \sqrt{(2/3) \cdot k} \approx (0.1 - 0.3)\, U_0$ (where the last relation comes from observation), thus their Reynolds number:

$$Re_0 = \frac{u_0\, \ell_0}{\nu} \approx \frac{U_0\, L_0}{\nu} \gg 1 \tag{2.13}$$

this suggests how at these scales the direct effect of molecular viscosity is negligible.

Thus, in order to achieve dissipation, unstable large eddies transfer energy to smaller ones. This process continues until the scale $\ell = \eta$ is reached, at this level the associated Reynolds number becomes sufficiently small, or in other words the molecular viscosity begins to be effective in energy dissipation.

The energy transferred to smaller scales can be evaluated as:

$$\frac{k_0}{\tau_0} \approx \frac{u_0^2}{(\ell_0/u_0)} = \frac{u_0^3}{\ell_0} \propto \frac{U_0^3}{L_0} \propto \mathcal{E} \tag{2.14}$$

the scaling of $\mathcal{E}$ with U_0^3/L_0, independently from ν (at the high Reynolds numbers considered) proves to be consistent with the experimental results for free shear flows [1].

2.4 Kolmogorov's theory

With the Kolmogorov theory, a better characterization of the involved scales in high-Reynolds flows was proposed. The theory was stated in three hypotheses:

- **Local isotropy hypothesis:** *At sufficiently high Reynolds numbers, the small-scales ($\ell \ll \ell_0$) turbulent motions are statistically isotropic.*

  While larger eddies show an anisotropic behavior, as a result of flow geometry and boundary conditions, scales of motion under a certain length value (small-scales) have statistical properties that are independent of the large, anisotropic scales. Consequently, they exhibit a universal behavior, in the sense that they appear similar in every high-Reynolds number flow.

- **First similarity hypothesis:** *In every turbulent flow at sufficiently high Reynolds number, the statistics of the small-scale motions have a universal form that is uniquely determined by ν and $\mathcal{E}$.*

  As a consequence of this hypothesis, starting from the ν and $\mathcal{E}$ parameters, it is possible to identify a particular set of scales (Kolmogorov scales) with specific length, velocity and time characteristics. This result is obtained by dimensional analysis.

$$\eta = \left(\frac{\nu^3}{\mathcal{E}}\right)^{1/4} \qquad u_\eta = (\nu\mathcal{E})^{1/4} \qquad \tau_\eta = \left(\frac{\nu}{\mathcal{E}}\right)^{1/2} \tag{2.15}$$

  These scales are considered the smallest active scales in the turbulent motion. As a support for this assertion it is possible to observe that the related Reynolds number is unity ($Re_\eta = \eta \cdot u_\eta/\nu = 1$), meaning that viscous forces effectively dominate over inertial forces, allowing the dissipation phenomena.

  It is possible to evaluate the entity of scales separation, considering the ratios of the smallest to largest scales. These ratios depend on the Reynolds number according to the following relations:

$$\frac{\ell_0}{\eta} \sim Re^{3/4} \qquad \frac{u_0}{u_\eta} \sim Re^{1/4} \qquad \frac{\tau_0}{\tau_\eta} \sim Re^{1/2} \tag{2.16}$$

- **Second similarity hypothesis:** *In every turbulent flow at sufficiently high Reynolds number, the statistics of the motions of scale ℓ in the range $\eta \ll \ell \ll \ell_0$ have a universal form that is uniquely determined by $\mathcal{E}$, independent of ν.*

  This intermediate range of scales is known as the inertial sub-range. In this region, the eddies are large enough that viscous effects are negligible (since $\ell \gg \eta$), but small enough that the large-scale anisotropic geometry is not felt (since $\ell \ll \ell_0$). Consequently, the energy cascade is dominated mainly by inertial forces. Again by dimensional analysis:

$$u_\ell = (\mathcal{E}\,\ell)^{1/3} \qquad \tau_\ell = \left(\frac{\ell^2}{\mathcal{E}}\right)^{1/3} \qquad Re_\ell = \left(\frac{\ell}{\eta}\right)^{4/3} \tag{2.17}$$

In summary, the energy input associated with the mean velocity field (for example as a result of convection) is in part transferred from one point of the domain to another by the flux term, another part is transferred down to the inertial subrange by the production term. This transfer to smaller and smaller scales continues until the Kolmogorov scales are reached, the effects of viscosity become dominant and the energy can be dissipated as heat.

2.5 Wall bounded flows

2.5.1 Inertial momentum balance

In the simple case of a uniform laminar flow above a smooth flat plate, the boundary layer develops in the flow direction. Its statistics vary primarily in the wall-normal y direction and they are independent on the span-wise z direction.

At this point it is useful to recall the RANS equations (2.6), developing their momentum x and y component, and neglecting the statistics variations in the z direction:

$$\frac{\partial U}{\partial t} + U\frac{\partial U}{\partial x} + V\frac{\partial U}{\partial y} = -\frac{1}{\rho}\frac{\partial P}{\partial x} + \nu\left(\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2}\right) - \left(\frac{\partial\langle u'u'\rangle}{\partial x} + \frac{\partial\langle u'v'\rangle}{\partial y}\right) \tag{2.18}$$

$$\frac{\partial V}{\partial t} + U\frac{\partial V}{\partial x} + V\frac{\partial V}{\partial y} = -\frac{1}{\rho}\frac{\partial P}{\partial y} + \nu\left(\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2}\right) - \left(\frac{\partial\langle u'v'\rangle}{\partial x} + \frac{\partial\langle v'v'\rangle}{\partial y}\right) \tag{2.19}$$

By applying the classical order of magnitude analysis originally proposed by Prandtl [2], based on the assumption that the boundary layer thickness is much smaller than the streamwise length scale, the equations above are further simplified. Moreover it is also possible to neglect the axial derivatives of the Reynolds stresses, that are small compared with lateral gradients [1].

For the cited assumptions, equation (2.18) becomes:

$$\frac{\partial U}{\partial t} + U\frac{\partial U}{\partial x} + V\frac{\partial U}{\partial y} = -\frac{1}{\rho}\frac{\partial P}{\partial x} + \nu\frac{\partial^2 U}{\partial y^2} - \frac{\partial\langle u'v'\rangle}{\partial y} \tag{2.20}$$

And can be rewritten as:

$$\underbrace{\frac{\partial U}{\partial t} + U\frac{\partial U}{\partial x} + V\frac{\partial U}{\partial y}}_{\text{Inertial terms}} = -\frac{1}{\rho}\frac{\partial P}{\partial x} + \frac{1}{\rho}\frac{\partial}{\partial y}\underbrace{\left[\mu\frac{\partial U}{\partial y} - \rho\langle u'v'\rangle\right]}_{\text{Total stress } \tau(y)} \tag{2.21}$$

The total shear stress $\tau(y)$ enclosed by brackets is composed by two contributions with different physical natures:

- **Viscous stress:** arises from the mean velocity gradients and is dominant in the immediate vicinity of the wall (inner region). It represents the molecular diffusion of momentum and is the only stress present in the laminar case.

- **Turbulent stress:** derives from the correlation of velocity fluctuations and represents the momentum transfer due to turbulent mixing. It prevails in the outer region, far from the wall, and vanishes in laminar conditions.

Under the same assumptions, equation (2.19) can be simplified and integrated as:

$$0 = -\frac{\partial P}{\partial y} - \rho \frac{\partial}{\partial y} \langle v'^2 \rangle \quad \Longrightarrow \quad 0 = \frac{\partial}{\partial y} \left(P + \rho \langle v'^2 \rangle \right) \tag{2.22}$$

In the particular case of flow over a flat plate considered, since at the wall and in the far free-stream region the term $\langle v'^2 \rangle$ cancels out, it results that the mean wall pressure $P_W(x)$ equals the free-stream pressure $P_E(x)$.

$$\left(P + \rho \langle v'^2 \rangle \right) = P_W(x) = P_E(x) \tag{2.23}$$

Equation (2.21) represents the *Inertial Momentum Balance*. The terms on the left-hand side account for the fluid inertia. These terms include both the unsteady acceleration and the spatial convection. Their presence indicates a non-equilibrium state: the pressure gradient and the total shear stress do not simply balance each other, and the local velocity profile is not determined uniquely by the instantaneous force balance, but depends on the flow evolution.

2.5.2 Equilibrium momentum balance

The previous case can be further simplified considering the example of a fully developed channel flow, characterized by more restrictive assumptions. The geometry consists of two infinite flat plates placed at a distance $2h$, with the flow driven by a constant mean pressure gradient along the streamwise direction x. In this configuration, the statistics are assumed to be independent of the streamwise coordinate x (except for the pressure), the spanwise coordinate z and the time t:

$$\frac{\partial}{\partial x} \langle ... \rangle = 0 \qquad \frac{\partial}{\partial z} \langle ... \rangle = 0 \qquad \frac{\partial}{\partial t} \langle ... \rangle = 0 \tag{2.24}$$

Due to the homogeneity in the x direction, the mass balance equation simplifies to $\partial V / \partial y = 0$. Considering the impermeability condition at the wall ($V|_{y=0} = 0$), it follows that the wall-normal mean velocity component V is null everywhere in the channel. Consequently, in equation (2.21), the inertial terms on the left-hand side vanish, and the x-momentum balance reduces to an equilibrium between the pressure forces and the gradient of the total stress:

$$0 = -\frac{1}{\rho} \frac{\partial P}{\partial x} + \frac{1}{\rho} \frac{\partial}{\partial y} \left[\mu \frac{\partial U}{\partial y} - \rho \langle u' v' \rangle \right] \tag{2.25}$$

Similarly with what already seen for equation (2.22), the term $P + \rho \langle v'^2 \rangle$ is constant along y, the term $\langle v'^2 \rangle$ cancels out at the wall, and its derivative in x is zero for the homogeneity

assumptions. Consequently:

$$\left(P + \rho \left\langle v'^2 \right\rangle\right) = P_W\left(x\right) \quad \Longrightarrow \quad \frac{\partial}{\partial x}\left(P + \rho \left\langle v'^2 \right\rangle\right) = \frac{\partial P}{\partial x} = \frac{\partial P_W}{\partial x} \tag{2.26}$$

And substituting the pressure gradient in equation (2.25):

$$\frac{dP_W}{dx} = \frac{d}{dy}\left[\mu \frac{dU}{dy} - \rho \langle u'v' \rangle\right] = \frac{d\tau}{dy} = const \tag{2.27}$$

While the pressure is a function of x, τ is a function of y so, in order for the equation (2.27) to be verified, their derivatives above must be equal to a constant value. This imply a linear behavior of the total shear stress along the y direction, described by:

$$\tau(y) = \tau_W\left(1 - \frac{y}{h}\right) \quad \Longrightarrow \quad \begin{cases} \tau(y=0) = \tau_W \\ \tau(y=2h) = -\tau_W \end{cases} \tag{2.28}$$

The data from DNS allows to evaluate the evolution of the two stress contributions, their plot over the half channel width is shown in Figure 2.1.

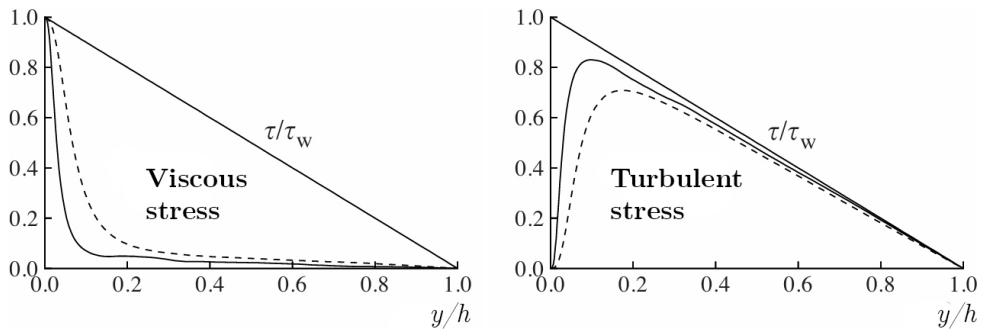


Figure 2.1: Contributions of normalized shear stress along half channel width, evaluated from DNS data [1] [3]. Dashed (solid) lines indicate lower (higher) Reynolds numbers.

Observing the data of the contributions, it is possible to identify two regions, and define for each one some important parameters:

- **Inner layer** ($y/h \ll 1$)**:** near the wall with a prevalence of the viscous effects. At increasing Reynolds number this region becomes progressively thinner. Its characteristic wall friction velocity and viscous length scale are defined as:

$$u_\tau = \sqrt{\frac{\tau_W}{\rho}} \qquad \delta_\nu = \frac{\nu}{u_\tau} \tag{2.29}$$

  Using these scales, the wall-normal coordinate and the mean velocity can be expressed in inner variables as:

$$U^+ = \frac{U}{u_\tau} \qquad y^+ = \frac{y}{\delta_\nu} = \frac{y\, u_\tau}{\nu} \tag{2.30}$$

It can be noticed that, to assume ($y/h \ll 1$) imply $\tau(y) \approx \tau_W$ and thus $dP_W/dx = d\tau/dy \approx 0$ (see equations (2.28) and (2.27)). This means that the gradients involved cannot be excessive in order for the formulation to be valid.

- **Outer layer** ($y/\delta_\nu \gg 1$)**:** where the turbulent stress component is dominant. The characteristic velocity and length scales are:

$$u_\tau = \sqrt{-\frac{h}{\rho}\frac{\partial P}{\partial x}} \qquad\qquad h \tag{2.31}$$

The ratio between the two characteristic length scales can be defined as the friction Reynolds number:

$$\frac{h}{\delta_\nu} = \frac{u_\tau}{\nu} = Re_\tau \tag{2.32}$$

Wall laws

Despite the previous separate definition of the two regions, the mean behavior of the flow varies with continuity moving away from the wall and so do the mathematical laws that describe the velocity profile:

- **Viscous sub-layer:** Inside the inner layer, in the close proximity of the wall, the flow is dominated by viscous effects. The linear relation is considered well verified:

$$U^+ = y^+ \qquad\qquad y^+ < 5 \tag{2.33}$$

- **Buffer layer:** In the transition region ($5 < y^+ < 30$), neither the linear or the logarithmic law provides an accurate description, as viscous and turbulent stresses are of comparable magnitude. Consequently, the velocity profile is described by semi-empirical functions (such as Van Driest damping or Spalding's law) that smoothly connect the viscous sub-layer to the logarithmic region.

- **Overlap region:** Moving towards the outer layer, the mean flow is described by the logarithmic law, still dependent on: the y^+ coordinate; the Von Karman constant ($\kappa = 0.41$); the integration constant ($B \approx 5.2$).

$$U^+ = \frac{1}{\kappa}\ln y^+ + B \qquad\qquad y^+ > 30 \tag{2.34}$$

It can be demonstrated that this law asymptotically transition to the defect law while approaching the outer layer. Then, the flow description is provided as a velocity defect with respect to the centerline velocity U_C. The defect law is dependent on: the y/h normalized coordinate; the Von Karman constant ($\kappa = 0.41$); the integration

constant B_1, specific for the type of flow.

$$\frac{U_C - U}{u_\tau} = -\frac{1}{\kappa}\ln\frac{y}{h} + B_1 \qquad y/h < 0.3 \tag{2.35}$$

The various layer just introduced are resumed in Figure 2.2.

Despite the previous relations have been derived with the assumption of a developed channel flow, multiple experiments verified the universality of such laws also for pipe flows and other equilibrium boundary layers, and as shown in Figure 2.3 they recover very accurately the experimental data

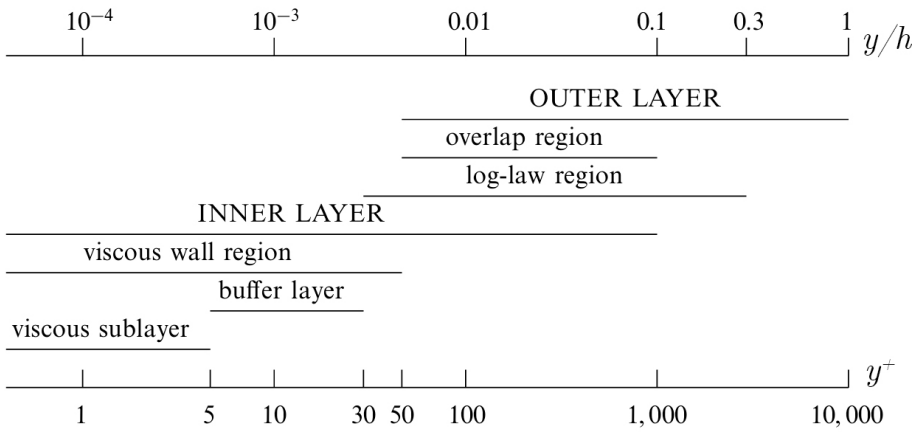


Figure 2.2: Wall regions for a turbulent channel flow [1].

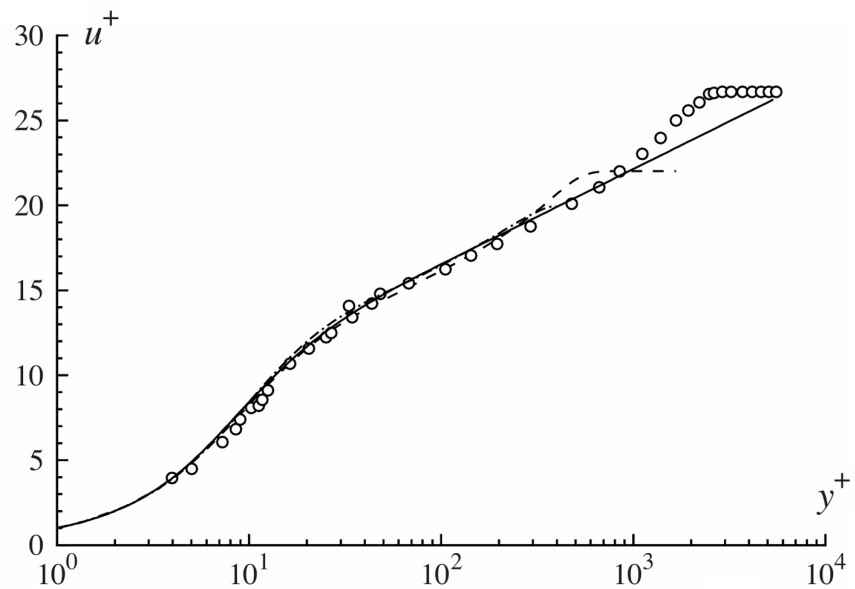


Figure 2.3: **Solid line**: linear and log laws (connected with the Van Driest function) [1]; **Circles** and **dashed line**: boundary layer above a flat plate, zero pressure gradient, data respectively from a hot-wire anemometer [4] and from DNS [5].

Chapter 3

Simulation techniques overview

The theoretical framework just introduced, highlights the wide range of scales of motion involved in the turbulent dynamics. In order to describe properly the fluid flow, it is necessary to take into account in the simulation all the space-time scales affecting the phenomena. This can result in an enormous amount of degrees of freedom, especially for high-Reynolds flows.

3.1 Direct Numerical Simulation

The most direct approach to perform a simulation, is choosing a space (grid) and time discretization able to represent all the space-time scales, included the smaller ones (Kolmogorov scales), in a non-stationary, three dimensional domain.

In terms of accuracy and level of description provided it is unrivalled at the point that these numerical experiments can be considered directly analogous to actual laboratory experiments [6] and does not need validation. However for the number of degrees of freedom involved its cost is extremely high and its applicability is limited to low or moderate Reynolds numbers [1].

The scale separation problem can be illustrated assuming to discretize a domain with dimensions $L \sim \ell_0$. To properly resolve the dissipative structures, the grid spacing Δ must be comparable to the Kolmogorov scale η. Applying the relations (2.16), the number of grid points required along each axis is on the order of $Re^{3/4}$ and in terms of time discretization, the required number of time-steps for a τ_0 interval would be on the order of $Re^{1/2}$. Overall, the computational cost of the simulation scales with the Reynolds number, following a $Re^{11/4} \approx Re^3$ relation.

As a consequence, this type of simulation for large Reynolds number, such as those related to application in the aeronautical field ($Re \sim 10^8$) requires computer resources very much

greater than currently available super computer capacities, and is therefore not practicable [7].

3.2 RANS-based simulation

An approach capable of reducing heavily the number of degrees of freedom to be computed is based on the Reynolds-Averaged Navier Stokes equations (2.6) previously described. In contrast to the direct resolution of all turbulent scales, such method focuses on the determination of the mean flow field. The contribution related to the fluctuation field (transfer of momentum due to turbulent fluctuations) appears in the equations as the Reynolds stress tensor $\langle \boldsymbol{u}'\boldsymbol{u}' \rangle$ that needs to be determined to close the system of equations.

The structure of the Reynolds stress tensor is:

$$\langle \boldsymbol{u}'\boldsymbol{u}' \rangle = \begin{bmatrix} \langle u'^2 \rangle & \langle u'v' \rangle & \langle u'w' \rangle \\ \langle v'u' \rangle & \langle v'^2 \rangle & \langle v'w' \rangle \\ \langle w'u' \rangle & \langle w'v' \rangle & \langle w'^2 \rangle \end{bmatrix} \tag{3.1}$$

it is constituted by nine components, but considering its symmetry, the independent terms are six. Recalling equation (2.7) that defines the turbulent kinetic energy, it is possible to observe that the trace of the tensor corresponds to twice the turbulent kinetic energy k.

From a computational perspective, the grid requirements for RANS simulations are significantly less stringent than for DNS. The mesh must only be fine enough to resolve the gradients of the mean flow and the geometry, rather than the smallest Kolmogorov scales. Consequently, the computational cost does not scale theoretically with the Reynolds number as strictly as in DNS, making RANS the industry standard for high-Reynolds number engineering applications [8], with best performances for attached flows.

However, this reduction in computational effort comes at the cost of physical fidelity. By modeling the entire spectrum of turbulence, RANS simulations rely heavily on empirical constants and assumptions (e.g., isotropy of turbulence) that are not always accurate in complex flow regimes. Another fundamental limitation arises from the time-averaging procedure itself, that filters the entire dynamic content of the flow. Consequently, they are unable to capture transient phenomena, which are critical in many aerodynamic applications. While the mean flow representation is often sufficient for preliminary design, it fails when the physical unsteadiness of the large scales is dominant. To address cases where global flow instabilities must be resolved, the Unsteady RANS (URANS) approach, reviewed in a later section, can be adopted.

While those characteristics are generally true, it should be noted that the advantages and limitations of RANS simulations are significantly dependent on the turbulence model of

choice. To have an overall understanding of their range of applicability, the main properties of those models will be summarized below.

To resolve the closure problem, varying levels of approximation can be adopted. Those approaches can be classified in the first place based on the order of resolved and modeled components (statistical moments).

3.2.1 First-order closures

The most common and simple class of approaches is represented by first-order closures. In this case the Reynolds stress tensor (that is a second moment), is determined through a relation with the mean flow velocity gradients (first moment). First-order closures can be further classified in linear, mostly based on the Boussinesq hypothesis, and non-linear eddy-viscosity formulations.

In linear formulations, the modeling of the Reynolds stress tensor can be split in two contributions: a isotropic one, related to k, and a deviatoric one. To model the latter, the Boussinesq hypothesis is introduced. This assumption relies on a direct analogy with a laminar velocity field, assuming that the turbulent shear stresses are linearly proportional to the mean strain rate tensor E_{ij} through a scalar coefficient known as the eddy viscosity, ν_T:

$$-\langle u_i' \, u_j' \rangle = -\frac{2}{3} \, k \, \delta_{ij} + 2 \, \nu_T \, E_{ij} \tag{3.2}$$

The fundamental advantage of this hypothesis is the significant reduction in the problem complexity. It transforms the task of computing six independent tensor components into the determination of a single scalar field, ν_T, that represent a function of the fluid local state of motion. However this approach introduces significant limitations. The assumption of local equilibrium between the mean strain tensor E_{ij} and the deviatoric part of the turbulent stress tensor and its consequent independence on fluid rigid rotations is often excessively restrictive for flows characterized by massive separation, strong curvature, or unsteady large-scale structures [8] [9].

With the development of more complex non-linear eddy viscosity models, the limitations discussed above are in part mitigated. They extend the relation between the Reynolds stress tensor and E_{ij}, involving the mean flow rotation tensor Ω_{ij} and including also non-linear quadratic and cubic terms [10].

First-order turbulence closures can be also classified by the number of governing equations used, those with a greater number of equation usually provide a fuller description of turbulence.

- Algebraic or zero-equations models such as the uniform turbulent viscosity one are characterized by extremely limited ranges of applicability.

- One-equation models are also very limited in possible application fields, an example is the Spalart-Allmaras model, designed for aerodynamic flows. The only real advantages of using one-equation model rather than a two-equation model are related with stability and simplicity of the numerical solution [9].

- Two-equation models such as $k - \mathcal{E}$, $k - \omega$ and their blended implementations like Shear Stress Transfer (SST) are the most widely employed and for their importance will be further analyzed below.

Model $k - \mathcal{E}$

The $k - \mathcal{E}$ model is based on the evaluation of the turbulent kinetic energy and dissipation rate. One of the main limitations is related to near wall behavior [8].

- In low Reynolds number turbulence models there is the need of some damping functions to limit k and $\mathcal{E}$ and an high grid resolution (first node at $y+ \leq 1$) in order to stay valid and resolve the viscous sub-layer.

- By contrast, high Reynolds number turbulence models don't need damping functions, they are less stringent in term of grid resolution (first node inside $10 \leq y+ \leq 100$) but require the implementation of wall functions, often based in the logarithmic wall-law, with the consequent loss of accuracy in case of a separated flow.

Since the method is based on the Boussinesq assumption, it inherits all the limitations related to the linear eddy viscosity modeling already discussed previously. A third aspect regards the (non) universality: the model requires *ad hoc* adjustments of its calibration constants based on the nature of the considered flow [1]. It also degrade in performance with adverse pressure gradients.

Model $k - \omega$

The $k - \omega$ model is the second most widely used two-equation model. The term ω is simply defined as: $\omega = \mathcal{E}/k$. It better performs for boundary layer flows and in presence of moderate adverse pressure gradients. The main drawback instead, is related to its sensibility to the arbitrary choice of ω boundary condition value [1] [9].

Model SST

The Shear Stress Transport (SST) model merges the $k - \omega$ formulation with a high Reynolds number $k - \mathcal{E}$ model, with the intent of combining the advantages of the two. The $k - \omega$ is used in the boundary layer since it doesn't require damping functions and it better performs in presence of adverse pressure gradients, instead $k - \mathcal{E}$ is employed in

wake regions of the boundary layer. The switching between the models is ensured by a blending function, dependent on wall proximity.

3.2.2 Second-order closures

Second-order closures, unlike first-order models that rely on the Boussinesq hypothesis represent a more complex and flexible class of approaches to determine the Reynolds stress tensor. Reynolds Stress Transport (RST) models allow to solve individual transport equations for the six independent component of the Reynolds stress tensor. This approach naturally include effects of streamline curvature, sudden changes in strain rate, secondary motions and the history effects [9]. With a similar accuracy, but a slightly lower level of description and complexity, the Algebraic Reynolds Stress (ARS) models employ only two transport equations, generally for the turbulent kinetic energy and for the dissipation rate. Despite their increased accuracy both RST and ARS models suffer numerical issues and they are more computationally demanding, so first-order models are more widely used in practice [8].

3.2.3 Unsteady RANS (URANS)

While in the stationary case the mean value can be considered as a time average, in the non-stationary case the ensemble average is used, more similarly to the theoretical description. Also, in such cases, the approach is to apply a RANS model in order to resolve some of the unsteady dynamics of the flow.

In the case of unsteady flows, this method can be substantially more successful in determining the mean flow than a steady RANS computation. By contrast, for the model assumption to be valid, a clear separation of scales is necessary, condition that is not always verified, so it is useful to distinguish between two cases. When unsteadiness is triggered by boundary conditions, scale separation generally holds and modeling assumptions remain valid. Instead, in presence of internal instabilities (e.g., bluff body flows), scale separation usually does not hold. Consequently, especially in this case the confidence in the result is limited without experimental validation [11].

3.3 Large Eddy Simulation

The Large Eddy Simulation (LES) represents an intermediate approach between the high-fidelity of DNS and the low computational cost of RANS. The idea behind this technique is based on the observation that turbulent flows contain a wide range of length and time scales, which play different roles in the flow dynamics.

The large-scale structures, often referred to as "eddies", contain the majority of the

turbulent kinetic energy and are responsible for the transport of momentum and conserved properties [8]. These scales are strictly dependent on the boundary conditions and the flow geometry, making them highly anisotropic and difficult to model. Conversely, the small-scale structures tend to be more isotropic and universal in their behavior, essentially acting as a sink for the kinetic energy transferred from the larger scales (energy cascade). Based on this distinction, LES applies a spatial filtering operation to the Navier-Stokes equations. The large, energy-carrying motions are explicitly resolved on the computational grid, while the effects of the unresolved scales, those smaller than the grid cut-off width, are modeled using Subgrid-Scale (SGS) models.

This strategy offers significant advantages over RANS approaches. Since the large-scale unsteady motions are resolved directly rather than averaged, LES is capable of capturing transient phenomena and flow instabilities, which are critical in aerodynamic flows involving separation or bluff bodies. The arbitrary aspects are confined only to the small scales, making the simulation less dependent on model calibration compared to RANS, where the entire turbulent spectrum is modeled.

However, these advantages come with a substantial computational cost. In terms of grid resolution requirements, LES places itself in an intermediate position between DNS and RANS, only requiring the resolution of the structures down to the cut-off width Δ. Consequently, the number of degrees of freedom is significantly lower than that of DNS, and substantially higher than RANS, where the mesh is only required to resolve the mean flow gradients. This requirement becomes particularly critical in wall-bounded flows, where the physical size of the energy-carrying eddies decreases proportionally to the distance from the wall [1]. In the so-called Wall-Resolved LES (WR-LES), the need to resolve these inner boundary layer structures leads to a computational cost that scales significantly with the Reynolds number ($Re^{13/7}$) [12] [13], with possible prohibitive costs. To overcome this limitation and apply LES to high-Reynolds number flows of engineering interest, Wall-Modeled LES (WM-LES) approaches are adopted to bypass the strict resolution requirements of the near-wall region.

Considering the central role of this methodology in the present work, and specifically the application of Wall-Modeled LES, this topic will be addressed in detail in the following chapter.

3.4 Hybrid RANS/LES Simulation

Another possible approach to drastically reduce the computational cost of simulating turbulent flows is to is to couple LES with the RANS models. The fundamental idea behind this strategy is to perform LES only in those regions where it is strictly necessary, typically where the flow is dominated by large-scale structures or massive separation.

Conversely, the RANS approach is employed in the remainder of the domain, such as in the boundary layer regions, where it is known to be reliable and computationally efficient.

Following the classification proposed by Fröhlich and von Terzi [11], these techniques are categorized into *unified* and *segregated* modeling.

Unified modeling utilizes the same velocity transport equation in the entire domain, while the transition between RANS and LES models is managed locally. The switching between different models can be performed with a *blending* function approach or by using a pure LES model in one part of the domain and a pure RANS model in the other, with the so called *interfacing* approach, this is the case for instance of Detached Eddy Simulations (DES). While this approach is user-friendly as it avoids explicit zonal boundaries, it suffers from the potential existence of a "gray area" at the interface.

Segregated modeling instead require the *a priori* decomposition of the domain into distinct regions. In each sub-domain an almost stand-alone RANS or LES computation are performed and they are coupled via boundary conditions at the interfaces. This allow each model to operate in its regime of validity, where it better performs. However, this method requires prior knowledge of the flow physics to define sub-domains and involves complex implementation for the coupling across interfaces.

Chapter 4

Mathematical framework

The review of simulation techniques presented in the previous chapter highlighted the capabilities and limitations of different turbulence modelling strategies. In the present work, the Large Eddy Simulation (LES) approach is selected to effectively capture the unsteady dynamics of the flow. This chapter details the mathematical and numerical framework of the solver, which has been specifically customized and further developed for the purposes of this study. As introduced previously, the LES approach relies on the application of a spatial filtering operation to the Navier-Stokes equations. This represents the starting point for the derivation of the governing equations, where the large scales are explicitly resolved, while the effects of the unresolved scales are modeled using Subgrid-Scale (SGS) models. Furthermore, the chapter provides a description of the advanced methodologies implemented to handle high-Reynolds number flows and complex geometries, specifically focusing on the Wall-Modelling strategies and the Immersed Boundary Method (IBM).

4.1 Filtering operation

Filtering is the operation that allows to mathematically separate the turbulent scales of motion. This decomposition is determined by a characteristic arbitrary cutoff width Δ_F. In the spectral domain, this procedure acts as a low-pass filter: it retains the low-frequency, large-scale contributions, while the high-frequency fluctuations associated with spatial scales smaller than Δ_F are cut off to be modeled. Considering a generic field, this decomposition can be mathematically represented as:

$$\phi\left(\boldsymbol{x},\, t\right) = \widetilde{\phi}\left(\boldsymbol{x},\, t\right) + \phi''\left(\boldsymbol{x},\, t\right) \tag{4.1}$$

Considering a homogeneous and isotropic case, where the filter acts independently of position and orientation, the general space-time filtering operation can be expressed as:

$$\widetilde{\phi}\left(\boldsymbol{x},\, t\right) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \phi\left(\boldsymbol{r},\, \tau\right) G\left(\boldsymbol{x} - \boldsymbol{r},\, t - \tau\right) d\tau\, d^3\boldsymbol{r} = G * \phi \tag{4.2}$$

where:

- ϕ represents the generic flow variable.
- $\widetilde{\phi}$ is the filtered generic flow variable.
- ϕ'' is the residual/sub-grid component.
- The double integral represent a temporal (one dimensional) and spatial (three dimensional) "scan" of the entire domain.
- G is the filter kernel, dependent on the choice of Δ_F.

Formally, this expression represents a convolution product between the physical quantity and the filter function. In simpler terms, this is a sort of weighted average operation of the values assumed by the field in the proximity of the considered point.

It should be noted that, despite the equation (4.2) represents a filtering operation in the general case (for both space and time), the imposition of a spatial filtering automatically induces an implicit time filtering, according to the Kolmogorov's relations (2.17), so it is possible to consider sufficient the spatial filtering alone [7].

4.1.1 Properties of the filtering operator

In order to derive the equations for LES from Navier-Stokes equations, the filtering operator is required to satisfy three fundamental properties:

1. **Conservation of constants:** The filter kernel must be normalized, such that the integral over the entire domain equals unity. This ensures that the filtering of a constant value returns the constant itself, preserving the global conservation of physical quantities.

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} G(\boldsymbol{x}, \tau)\, d\tau\, d^3\boldsymbol{x} = 1 \tag{4.3}$$

2. **Linearity:** The filtering operation is linear, meaning that the filter of a sum is equal to the sum of the filtered quantities.

$$\widetilde{\phi + \psi} = \widetilde{\phi} + \widetilde{\psi} \tag{4.4}$$

3. **Commutation with differentiation:** The result is independent of the sequence in which differentiation and filtering are performed.

$$\frac{\partial \widetilde{\phi}}{\partial s} = \widetilde{\frac{\partial \phi}{\partial s}}, \qquad s = \boldsymbol{x}, t \tag{4.5}$$

It is important to observe that a filter satisfying these properties differs substantially from the Reynolds averaging operator used in RANS. As a consequence:

$$\widetilde{\widetilde{\phi}} \neq \widetilde{\phi} \qquad \widetilde{\phi''} \neq 0 \tag{4.6}$$

This implies that applying the filter twice produces a result different from a single application and the filtered fluctuation is not necessarily zero.

4.1.2 Explicit and implicit filtering

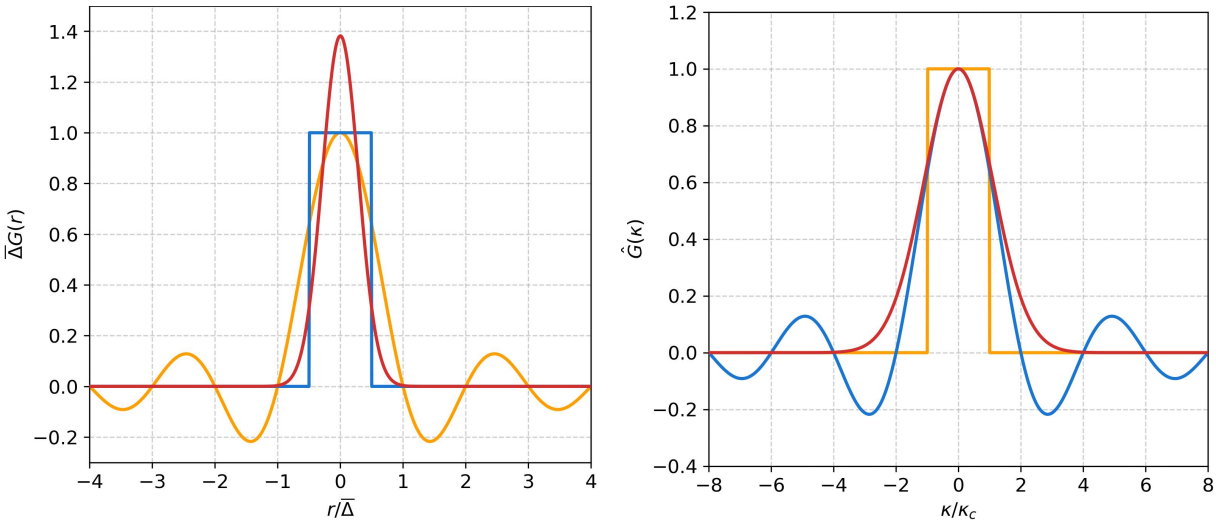
It is important to distinguish between explicit and implicit filtering.

The explicit filtering approach involves the application of a distinct mathematical operator to the flow field, with a defined characteristic length Δ_F, independent on the grid resolution Δ. The three most common explicit filters are (Figure 4.1):

- **Box or top-hat filter:** performs a local spatial average of the flow variables over a defined control volume. Its oscillating transfer function causes the introduction of high wavenumber noise.
- **Gaussian filter:** is based on a spatial Gaussian distribution. It has smooth characteristics in both physical and spectral spaces preventing the generation of numerical oscillations.
- **Sharp spectral cutoff filter:** eliminates all Fourier modes exceeding a specific cutoff wavenumber. Although it provides a definitive separation of scales, its non-locality in physical space can induce significant oscillations.

This explicit approach allows decouple the filter length and the grid resolution obtaining a better control on which scales are resolved and which are modeled. This has important implications in term of stability control and allows to perform mesh convergence analysis varying the grid resolution alone.

Conversely, the implicit filtering approach relies on the computational grid and the discretization scheme to act as the filter. In this case, the filter width is coupled with the local mesh resolution ($\Delta_F = \Delta$), assuming that the grid itself defines the cutoff frequency without the need for an external mathematical operator. This allows to save a relevant part of the computational cost.



(a) Physical space representation (b) Spectral space representation

Figure 4.1: One-dimensional (for simplicity) comparison of the three primary filter kernels. **Blue** for the box (top-hat) filter, **red** for the Gaussian filter, and **yellow** for the sharp spectral cutoff filter.

4.2 Governing equations

Applying the filtering operation to the incompressible Navier-Stokes equations, the governing equations for LES are obtained:

$$\begin{cases} \dfrac{\partial \widetilde{u}_i}{\partial x_i} = 0 \\ \dfrac{\partial \widetilde{u}_i}{\partial t} + \widetilde{u}_j \dfrac{\partial \widetilde{u}_i}{\partial x_j} = -\dfrac{1}{\rho}\dfrac{\partial \widetilde{p}}{\partial x_i} + \dfrac{\partial}{\partial x_j}(2\,\nu\,\widetilde{e}_{ij}) - \dfrac{\partial}{\partial x_j}(\widetilde{u_i u_j} - \widetilde{u}_i\,\widetilde{u}_j) + f_i \end{cases} \tag{4.7}$$

where $\widetilde{e}_{ij}$ represents the resolved rate-of-strain tensor and τ_{ij}^R is the residual (or sub-grid-scale) stress tensor:

$$\widetilde{e}_{ij} = \frac{1}{2}\left(\frac{\partial \widetilde{u}_i}{\partial x_j} + \frac{\partial \widetilde{u}_j}{\partial x_i}\right) \qquad \tau_{ij}^R = (\widetilde{u_i u_j} - \widetilde{u}_i\,\widetilde{u}_j) \tag{4.8}$$

The residual stress tensor τ_{ij}^R represents the effect of the unresolved scales on the resolved field and, being symmetric ($\tau_{ij}^R = \tau_{ji}^R$), it introduces six independent components.

The system consists of only 4 equations (1 continuity and 3 momentum equations), while the dependent variables to be determined are 10: the three filtered velocity components $\widetilde{u}_i$, the filtered pressure $\widetilde{p}$, and the six independent components of the residual stress tensor. Since the number of unknowns (10) exceeds the number of available equations (4), the system is mathematically unclosed. Therefore, a closure model is required to express the unknown stresses τ_{ij}^R in terms of the resolved flow variables.

4.2.1 Sub-Grid Scales modelling

The modelling of the coupling term between resolved and sub-grid scales has to be performed on the basis of the resolved scales alone. As explained by Sagaut [7] this can be done with two approaches:

- **Structural modelling:** Allows for the modelling of the entire tensor τ_{ij}^R, from the field $\widetilde{\boldsymbol{u}}$, requiring a relevant knowledge of the sub-grid scales structures. This is feasible when small scales shows an universal structure or a clear correlation with the structure of the resolved field.

- **Functional modelling:** Consist in the modelling of the action of the tensor τ_{ij}^R, without determining the tensor itself. This require an understanding of the nature of the interaction and an effect of the small scales on the large universal in character.

Smagorinsky model

As most of the strategies for functional modelling, the Smagorinsky approach models the sub-grid scale tensor τ_{ij}^R, splitting it in a isotropic and a deviatoric contribution:

$$-\tau_{ij}^R = -\frac{2}{3}\, k_R\, \delta_{ij} + 2\, \nu_R\, \widetilde{e}_{ij} \tag{4.9}$$

The first term is related to the sub-grid kinetic energy k_R, while the second one is modeled as a term linearly proportional to the deformation tensor of the resolved field $\widetilde{e}_{ij}$ through the sub-grid scale eddy viscosity coefficient ν_R.

Substituting this closure into the filtered momentum equation (4.7), the isotropic part of the residual stress is absorbed into the pressure term. Defining a modified pressure $\widetilde{p}$ (which now includes the trace of the sub-grid stress tensor) and grouping the molecular and eddy viscosities, the governing equation for the large scales becomes:

$$\frac{\partial\, \widetilde{u}_i}{\partial\, t} + \widetilde{u}_j\, \frac{\partial\, \widetilde{u}_i}{\partial\, x_j} = -\frac{1}{\rho}\, \frac{\partial\, \widetilde{p}}{\partial\, x_i} + \frac{\partial}{\partial x_j}\left[(\nu + \nu_R)\left(\frac{\partial \widetilde{u}_i}{\partial x_j} + \frac{\partial \widetilde{u}_j}{\partial x_i}\right)\right] + f_i \tag{4.10}$$

Exploiting the incompressibility constraint ($\nabla \cdot \widetilde{\boldsymbol{u}} = 0$) to simplify the viscous term, it is possible to retrieve the formulation implemented in the numerical solver. The equation can be rewritten in vector notation as:

$$\frac{\partial\, \widetilde{\boldsymbol{u}}}{\partial t} + \widetilde{\boldsymbol{u}} \cdot \nabla \widetilde{\boldsymbol{u}} = -\frac{1}{\rho}\, \nabla \widetilde{p} + \nabla \cdot \left[(\nu + \nu_R)\, \left(\nabla \widetilde{\boldsymbol{u}} + (\nabla \widetilde{\boldsymbol{u}}\,)^T\right)\right] + \boldsymbol{f} \tag{4.11}$$

In the Smagorinsky model the ν_R scalar is defined as:

$$\nu_R = (C_S\, \Delta_F)^2\, |\widetilde{e}_{ij}| \qquad |\widetilde{e}_{ij}| := \sqrt{2\, \widetilde{e}_{ij}\, \widetilde{e}_{ij}} \tag{4.12}$$

where Δ_F is the sub-grid characteristic length scale (or filter size), $|\widetilde{e}_{ij}|$ is a characteristic sub-grid velocity (defined above) and C_S is a tuning constant.

The idea behind the definition of the eddy viscosity ν_R is to emulate the energy transfer between resolved scales and sub-grid ones, by introducing an artificial dissipation at the Δ_F scale.

Model limitations While the model satisfies some useful properties, like its independence on coordinate translation and rotation and its easy implementation on any kind of computational grid, its predictions suffer from several aspects:

1. **Insensitivity to rotational flow structures:** To understand the main limitations of the Smagorinsky model, it is useful to recall the formulation of the velocity gradient tensor $\nabla \widetilde{\boldsymbol{u}}$:

$$\nabla \widetilde{\mathbf{u}} = \frac{\partial \widetilde{u}_i}{\partial x_j} = \underbrace{\frac{1}{2}\left(\frac{\partial \widetilde{u}_i}{\partial x_j} + \frac{\partial \widetilde{u}_j}{\partial x_i}\right)}_{\widetilde{e}_{ij}\ \text{(Strain rate)}} + \underbrace{\frac{1}{2}\left(\frac{\partial \widetilde{u}_i}{\partial x_j} - \frac{\partial \widetilde{u}_j}{\partial x_i}\right)}_{\widetilde{\omega}_{ij}\ \text{(Rotation rate)}} \tag{4.13}$$

   The studies of Wray and Hunt [14] of the properties of isotropic turbulence structures, based on DNS analyses, showed that while kinetic energy is concentrated in the streams, the dissipation phenomenon occurs in:

   - Eddies: where vorticity is dominant over irrotational strain ($|\widetilde{\boldsymbol{\omega}}| \gg |\widetilde{\boldsymbol{e}}|$).
   - Convergence zones: where the prevalent term is the irrotational strain ($|\widetilde{\boldsymbol{e}}| \gg |\widetilde{\boldsymbol{\omega}}|$).

   Since the Smagorinsky formulation takes into account the $\widetilde{e_{ij}}$ term alone, it is sensitive only to convergence zones, and it fails to correctly model regions where vorticity is dominant.

2. **Inadequacy for pure shear and transitional flows:** In the case of pure shear flows, the large values of the $|\widetilde{e}_{ij}|$ invariant causes an overestimation of the energy to be virtually dissipated, preventing the growth of instabilities and consequently failing to reproduce the laminar-to-turbulent transition.

3. **Incorrect near wall behavior:** By construction, the Smagorinsky formulation yields a non-zero eddy viscosity ν_t whenever a velocity gradient exists. However, turbulent fluctuations are naturally damped near solid boundaries, implying that ν_t should disappear at the wall. A common approach to correct this behavior is the use of Van Driest damping functions. While this approach can improve the results, this is an ad hoc modification that depends on the distance to the wall (y), making it difficult to implement for complex geometries. Furthermore, this damping leads to

an eddy-viscosity scaling of $O(y^2)$, which is inconsistent with the correct theoretical behavior of $O(y^3)$ [15].

WALE model

The WALE model addresses the limitations of Smagorinsky one, while maintaining its advantageous properties of independence of coordinate transformations and grid topology choice.

Considering the traceless symmetric part of the square of the velocity gradient tensor:

$$\mathcal{S}_{ij} = \frac{1}{2}(\,\widetilde{g}_{ij}^2 + \widetilde{g}_{ji}^2\,) - \frac{1}{3}\,\delta_{ij}\,\widetilde{g}_{kk}^2 \tag{4.14}$$

where the terms:

$$\widetilde{g}_{ij} = \frac{\partial\,\widetilde{u}_i}{\partial\,x_j} = \widetilde{e}_{ij} + \widetilde{\omega}_{ij} \qquad \widetilde{g}_{ij}^2 = \frac{\partial\,\widetilde{u}_i}{\partial\,x_k}\,\frac{\partial\,\widetilde{u}_k}{\partial\,x_j} = (\widetilde{e}_{ik} + \widetilde{\omega}_{ik})(\widetilde{e}_{kj} + \widetilde{\omega}_{kj}) \tag{4.15}$$

Developing the square of the tensor defined in equation (4.14), the scalar $\mathcal{S}_{ij}\,\mathcal{S}_{ij}$ is obtained:

$$\mathcal{S}_{ij}\,\mathcal{S}_{ij} = \frac{1}{6}\,(\widetilde{e}^2\,\widetilde{e}^2 + \widetilde{\omega}^2\,\widetilde{\omega}^2) + \frac{2}{3}\,\widetilde{e}^2\,\widetilde{\omega}^2 + 2\,IV_{e\omega} \tag{4.16}$$

In which the participating components are:

- The squared magnitude of the resolved strain rate: $\widetilde{e}^2 = \widetilde{e}_{ij}\,\widetilde{e}_{ij}$
- The squared magnitude of the resolved rotation rate: $\widetilde{\omega}^2 = \widetilde{\omega}_{ij}\,\widetilde{\omega}_{ij}$
- A strain-rotation interaction term: $IV_{e\omega} = \widetilde{e}_{ik}\,\widetilde{e}_{kj}\,\widetilde{\omega}_{jl}\,\widetilde{\omega}_{li}$.

The formulated invariant (4.16), is capable of detecting turbulent structures dominated by either strain, rotation, or both. Moreover, in the case of pure shear (e.g., laminar Poiseuille flow), the strain and rotation magnitudes balance each other ($\widetilde{e}^2 \approx \widetilde{\omega}^2$), causing the invariant to vanish. This property ensures that the sub-grid dissipation is naturally zero in laminar wall-bounded regions, allowing the development of instabilities and thus enabling the correct prediction of the laminar-to-turbulent transition.

At this point some corrections are implemented to ensure the correct $O(y^3)$ near wall scaling, dimensional coherency and to avoid numerical instabilities related to the cancellation of the denominator. Finally, the expression for the WALE model is the following:

$$\nu_R = (C_W\Delta_F)^2\,\frac{(\mathcal{S}_{ij}\,\mathcal{S}_{ij})^{3/2}}{(\widetilde{e}_{ij}\,\widetilde{e}_{ij})^{5/2} + (\mathcal{S}_{ij}\,\mathcal{S}_{ij})^{5/4}} \tag{4.17}$$

Chapter 5

Numerical framework

5.1 CaNS solver

The numerical framework used is based on *CaNS (Canonical Navier-Stokes)*, an open-source solver originally developed by Costa [16] for massively parallel Direct Numerical Simulations (DNS) of canonical flows. The baseline code provides a highly efficient architecture for solving the incompressible Navier-Stokes equations on structured Cartesian grids. However, to address the high Reynolds number flows and complex geometries characterizing the present research, the original solver required specific adaptations. Consequently, the code was modified to operate as a Wall-Modeled Large Eddy Simulation (WM-LES) solver coupled with an Immersed Boundary Method (IBM).

The algorithm solves the governing filtered Navier-Stokes equations presented in equation (4.11) with unit density. To describe the time-integration scheme, it is convenient to recast the momentum equation by grouping the convective (or advective) and diffusive terms into a single operator, denoted here as $\mathcal{AD}(\widetilde{\boldsymbol{u}})$:

$$\mathcal{AD}(\widetilde{\boldsymbol{u}}) = -\nabla\left(\widetilde{\boldsymbol{u}}\,\widetilde{\boldsymbol{u}}\right) + \nabla\cdot\left[\left(\nu+\nu_R\right)\left(\nabla\widetilde{\boldsymbol{u}} + \left(\nabla\widetilde{\boldsymbol{u}}\right)^T\right)\right] \tag{5.1}$$

Using this notation, the system of equations solved by the algorithm simplifies to:

$$\begin{cases} \nabla\cdot\widetilde{\boldsymbol{u}} = 0 \\ \dfrac{\partial\,\widetilde{\boldsymbol{u}}}{\partial t} = \mathcal{AD}(\widetilde{\boldsymbol{u}}) - \nabla\widetilde{p} + \boldsymbol{f} \end{cases} \tag{5.2}$$

The core of the solver workflow is a three-stage Runge-Kutta (RK3) scheme for time integration, with a finite-volume pressure-correction method for the coupling of the mass and momentum Navier-Stokes equations for each sub-step.

5.1.1 Time integration - RK3

Runge–Kutta schemes perform time advancement by evaluating the solution at step $n+1$ via a sequence of intermediate stages. Considering a generic problem and a standard three-stage Runge-Kutta method:

$$\begin{cases} \boldsymbol{u}_1 = \boldsymbol{u}_n \\ \boldsymbol{u}_2 = \boldsymbol{u}_n + \Delta t\, a_{21}\ (RHS_1) \\ \boldsymbol{u}_3 = \boldsymbol{u}_n + \Delta t\, \{a_{31}\ (RHS_1) + a_{32}\ (RHS_2)\} \\ \boldsymbol{u}_{n+1} = \boldsymbol{u}_n + \Delta t\, \sum_{k=1}^{3} b_k\ (RHS_k) \end{cases} \tag{5.3}$$

The last equation yields the $n+1$ field with a weighted average of the intermediate solutions found in each k-sub-step.

The use of multiple sub-steps allows for a significant increase in accuracy and stability, but at the same time the storage of the arrays for each stage causes an excessive memory consumption. For this reason, low-storage versions of the scheme are usually implemented. Unlike the standard formulation, the low-storage scheme updates the solution recursively "in-place":

$$\boldsymbol{u}_{k+1} = \boldsymbol{u}_k + \Delta t\, \{\alpha_k\ (RHS_k) + \beta_k\ (RHS_{k-1})\} \tag{5.4}$$

it is worth noting that for the first sub-step ($k = 1$), the coefficient β_1 is zero, reducing the dependence solely to the initial state. This recursive structure allows the solver to operate using a minimal number of memory registers, by overwriting the solution array at each stage.

Using the time integration scheme as is, to solve the (5.2) momentum equation alone is not possible, since both the mass and momentum conservation have to be imposed. In order to couple the four equations, and ensure a divergence-free velocity, the solver employs a pressure-correction algorithm, nested in each sub-step of the RK3 cycle.

5.1.2 Pressure-correction

The algorithm can be summarized as follows:

1. **Prediction:** The procedure begins with the computation of a predicted velocity field, denoted as $\boldsymbol{u}^*$, which is not necessarily divergence-free. This intermediate field is obtained by advancing the momentum equation using the convective and diffusive terms $\mathcal{AD}$ evaluated at current and previous stages, along with the pressure gradient and body forces term. Consistent with the low-storage RK3 scheme described in

equation (5.4), the prediction step is formulated as:

$$\boldsymbol{u}^* = \boldsymbol{u}_k + \Delta t \left(\alpha_k \, \mathcal{AD}_k + \beta_k \, \mathcal{AD}_{k-1} + \gamma_k \left(-\nabla p_{k-1/2} + \boldsymbol{f} \right) \right) \tag{5.5}$$

It is crucial to observe the time-centering of the pressure term. In this formulation, the pressure is staggered in time with respect to the velocity field. The term $\nabla p_{k-1/2}$ represents the pressure gradient evaluated at the mid-point of the previous interval.

The discrete form of Equation (5.5) is implemented as follows:

src/rk.f90

```
factor1 = rkpar(1) * dt
factor2 = rkpar(2) * dt
factor12 = factor1 + factor2
...
u(i,j,k) = u(i,j,k) + factor1 * dudtrk(i,j,k)
        + factor2 * dudtrko(i,j,k)
        + factor12 * (bforce(1) - dli(1) * (p(i+1,j,k) - p(i,j,k)))
v(i,j,k) = v(i,j,k) + factor1 * dvdtrk(i,j,k)
        + factor2 * dvdtrko(i,j,k)
        + factor12 * (bforce(2) - dli(2) * (p(i,j+1,k) - p(i,j,k)))
w(i,j,k) = w(i,j,k) + factor1 * dwdtrk(i,j,k)
        + factor2 * dwdtrko(i,j,k)
        + factor12 * (bforce(3) - dzci(k) * (p(i,j,k+1) - p(i,j,k)))
```

where, for instance, `dudtrk` and `dudtrko` are the the x-direction convective/diffusive terms at the current and previous (older) sub-steps.

2. **Poisson equation resolution:** Since, as anticipated, the evaluated $\boldsymbol{u}^*$ does not satisfy the continuity equation, it requires a correction. Such correction is enforced with a proper pressure correction field $p^* = p_{k+1/2} - p_{k-1/2}$, to obtain the final divergence free velocity of the current sub-step $\boldsymbol{u}_{k+1}$:

$$\boldsymbol{u}_{k+1} - \boldsymbol{u}^* = -\Delta t \, \gamma_k \, \nabla p^* \tag{5.6}$$

Applying the divergence to both sides:

$$0 - \nabla \cdot \boldsymbol{u}^* = -\Delta t \, \gamma_k \, \nabla^2 p^* \qquad \Longrightarrow \qquad \nabla^2 p^* = \frac{\nabla \cdot \boldsymbol{u}^*}{\gamma_k \, \Delta t} \tag{5.7}$$

The solution of the Poisson equation is usually the most computationally intensive

step of a solver. In *CaNS*, a Fast Fourier Transform (FFT) based method is employed, applied along the x and y domain dimensions [16].

3. **Velocity correction:** The divergence-free velocity at the subsequent step $\boldsymbol{u}_{k+1}$ can be retrieved by reconsidering equation (5.6), that becomes:

$$\boldsymbol{u}_{k+1} = \boldsymbol{u}^* - \Delta t\, \gamma_k\, \nabla p^* \tag{5.8}$$

4. **Pressure update:** The pressure field at step $k + 1/2$ is obtained by summation:

$$p_{k+1/2} = p_{k-1/2} + p^* \tag{5.9}$$

5.1.3 Finite Volume Method

The previous sections described the time integration scheme and the pressure correction algorithm. To complete the description, it is necessary to define how the spatial terms of equation (5.2) are discretized.

The solver uses a standard second-order Finite Volume Method (FVM) on a structured Cartesian grid. This approach is based on the integral form of the conservation laws applied to elementary control volumes. The main advantage of this method is that it naturally ensures the conservation of mass and momentum fluxes across the cell faces. For this reason the method is often implemented for its ability to compute flow fields which present discontinuities (for instance shocks, slip lines), namely weak solutions.

To obtain the integral form of the governing equations, the Gauss's Divergence Theorem is applied. This theorem allows to transform the volume integral into a surface integral, which can be subsequently approximated by a summation over the faces of the discrete control volume. Considering a generic field, for instance $\boldsymbol{u}$, the continuous and discrete forms of the theorem are recalled below:

$$\int_\Omega (\nabla \cdot \widetilde{\boldsymbol{u}})\, d\Omega = \int_S \widetilde{\boldsymbol{u}} \cdot d\boldsymbol{S} \quad \Longrightarrow \quad \sum_f \boldsymbol{u_f} \cdot \boldsymbol{S}_f = \sum_f \boldsymbol{F}_f^{vel} \cdot \boldsymbol{S}_f \tag{5.10}$$

where, in this example, $\boldsymbol{S}$ is the surface vector, $(\boldsymbol{u_f} \cdot \boldsymbol{S}_f)$ is the flux contribution across the face f, and $\boldsymbol{F}_f^{vel} \equiv \boldsymbol{u_f}$ is the (velocity) flux density again across the face f. Note that the pedix f act as the position of evaluation of fluid quantities.

Integrating (5.2) in a cell control volume Ω and applying the divergence theorem:

$$\int_\Omega \frac{\partial \widetilde{\boldsymbol{u}}}{\partial t}\, d\Omega = -\int_S (\nu + \nu_R) \left(\nabla \widetilde{\boldsymbol{u}} + (\nabla \widetilde{\boldsymbol{u}})^T \right) dS - \int_S (\widetilde{p}\, \boldsymbol{I})\, dS + \int_\Omega \boldsymbol{f}\, d\Omega \tag{5.11}$$

This equation can be rewritten in its semi-discretized form, considering $\Omega = \Delta x\,\Delta y\,\Delta z$, as:

$$
\begin{aligned}
(\Delta x\,\Delta y\,\Delta z)\,\frac{\partial\,\boldsymbol{u}}{\partial t} = &-\sum_f (\boldsymbol{u}\,\boldsymbol{u})_f \cdot \boldsymbol{S}_f \\
&+\sum_f \left[(\nu+\nu_R)\,\left(\nabla\boldsymbol{u} + (\nabla\boldsymbol{u})^T\right)\right]_f \cdot \boldsymbol{S}_f \\
&-\sum_f (p\,\boldsymbol{I}) \cdot \boldsymbol{S}_f \\
&+(\Delta x\,\Delta y\,\Delta z)\;\boldsymbol{f}
\end{aligned}
\tag{5.12}
$$

That is, in indexed notation:

$$
\begin{aligned}
(\Delta x\,\Delta y\,\Delta z)\,\frac{\partial\,u_i}{\partial t} = &-\sum_f (u_i\,u_j)_f\,S_{f,j} \\
&+\sum_f \left[(\nu+\nu_R)\,(\nabla_j\,u_i + \nabla_i\,u_j)\right]_f\,S_{f,j} \\
&-\sum_f p_f\,S_{f,i} \\
&+(\Delta x\,\Delta y\,\Delta z)\,f_i
\end{aligned}
\tag{5.13}
$$

The flux density terms appearing can be identified. Specifically, the convective and viscous flux density tensors are defined as:

$$
\begin{aligned}
F^{conv}_{i,j,f} &= (u_i\,u_j)_f \\
F^{visc}_{i,j,f} &= \left[(\nu+\nu_R)\,(\nabla_j u_i + \nabla_i u_j)\right]_f
\end{aligned}
\tag{5.14}
$$

where the index i express the direction in which the momentum conservation equation is considered, j indicates the flux vector component direction and f the position in which the tensor is evaluated. Since the considered grid is Cartesian, the considered face f already determines the considered flux direction, so that the j index is redundant and can be suppressed.

Finally, to bridge the theoretical formulation with the numerical implementation, the specific notation adopted in the code is introduced. A distinction is made between the generic tensor indices and the specific spatial directions: while the lowercase subscripts (i, j, k) act as placeholders, the uppercase letters I, J, K represent the considered direction corresponding to the x, y, and z axes, respectively. Furthermore, the control volume faces are identified by their orientation relative to the cell center, using P (Plus) for the positive direction and M (Minus) for the negative direction.

For instance, considering the momentum balance (5.13) in the $y \equiv J$ direction:

$$
\begin{aligned}
(\Delta x\,\Delta y\,\Delta z)\,\frac{\partial\, v}{\partial t} = &-(F^{conv}_{J,IP} - F^{conv}_{J,IM})(\Delta y\,\Delta z) - (F^{conv}_{J,JP} - F^{conv}_{J,JM})(\Delta x\,\Delta z) \\
&-(F^{conv}_{J,KP} - F^{conv}_{J,KM})(\Delta x\,\Delta y) \\
&+(F^{visc}_{J,IP} - F^{visc}_{J,IM})(\Delta y\,\Delta z) + (F^{visc}_{J,JP} - F^{visc}_{J,JM})(\Delta x\,\Delta z) \\
&+(F^{visc}_{J,KP} - F^{visc}_{J,KM})(\Delta x\,\Delta y) \\
&-(p_{JP} - p_{JM})(\Delta x\,\Delta z) + (\Delta x\,\Delta y\,\Delta z)\,f_J
\end{aligned}
\tag{5.15}
$$

And simplifying the volume:

$$
\begin{aligned}
\frac{\partial v}{\partial t} = &-\frac{F^{\text{conv}}_{J,IP} - F^{\text{conv}}_{J,IM}}{\Delta x} - \frac{F^{\text{conv}}_{J,JP} - F^{\text{conv}}_{J,JM}}{\Delta y} - \frac{F^{\text{conv}}_{J,KP} - F^{\text{conv}}_{J,KM}}{\Delta z} \\
&+\frac{F^{\text{visc}}_{J,IP} - F^{\text{visc}}_{J,IM}}{\Delta x} + \frac{F^{\text{visc}}_{J,JP} - F^{\text{visc}}_{J,JM}}{\Delta y} + \frac{F^{\text{visc}}_{J,KP} - F^{\text{visc}}_{J,KM}}{\Delta z} \\
&-\frac{p_{JP} - p_{JM}}{\Delta y} + f_J
\end{aligned}
\tag{5.16}
$$

5.2 Immersed Boundary Method

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